

IDSPACE: A Novel Document Generator for Reliable Evaluation of Digital Identity Verification Systems

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Abstract—As services move online, institutions such as banks, lenders, and governments must verify the identities of remote users. Fraud detection tools are widely available, but evaluating and fine-tuning them remains difficult because identity documents are sensitive and therefore scarce. Synthetic data generation offers a path forward, and the demand is clear: our prior work in this area has been downloaded over 12,000 times (aggregated from eight parts). We introduce IDSPACE, extending this line of research in three directions. First, we propose model-guided Bayesian optimization, which tunes generation parameters to maximize both visual similarity and prediction consistency with target domain models given only a few samples from a target domain. Second, we decouple user-specified metadata (demographics, fraud patterns, capture device) from automatically tuned control parameters (font styles, noise levels, image quality), allowing users to configure evaluations without low-level expertise. Third, we expand beyond template images to support scanned and mobile-captured documents. Experiments show IDSPACE improves evaluation consistency by 15 – 45% over baselines, including CycleGAN, diffusion inpainting, and non-guided optimization, using only a few real samples, while improving training accuracy by up to 9% and SSIM similarity with the target domain by 10%. We also released a new dataset consisting of 359,240 high-quality synthetic documents across ten European ID types.

I. INTRODUCTION

The surge in digital platforms offering remote identity verification has raised concerns about forged identity documents, such as passports, driver’s licenses, and ID cards. Accurate detection of fraudulent identity documents is crucial for reducing authentication risks across various sectors, including finance, healthcare, travel, retail, government, and gambling [1]. Therefore, it is important to evaluate the accuracy of fraud detection capabilities before deploying digital identity verification services in these application scenarios. However, due to the sensitivity of personal information in these documents, a significant **data scarcity** challenge arises in assembling comprehensive real-world datasets for flexible and reliable evaluation of identity verification systems.

Motivating Example. Between August 2023 and April 2024, the US General Services Administration conducted a large-scale study evaluating the fairness of commercial remote identity verification services across demographic groups [2]. They recruited 3,991 participants from five racial and ethnic groups, who captured and uploaded images of their identity documents and selfies using mobile devices. This study illustrates the fundamental challenges of evaluating identity verification systems with real data: recruitment costs scale

linearly with coverage requirements, making it expensive to achieve statistical significance across demographic subgroups and document types; image quality variance is uncontrolled as participants use diverse devices under varying conditions; and demographic balance is difficult to achieve, with minority groups, precisely the populations where fairness audits matter most, often underrepresented. A synthetic generation framework that preserves evaluation validity while offering precise control over document characteristics would address both limitations.

Synthetic data generation offers a path forward. Existing synthetic identity document generation falls into three main categories, and they all suffer from significant limitations: **(Category 1) Manual generation** of identity document datasets, such as MIDV-500 [3] and MIDV-2020 [4], is expensive and labor-intensive, leading to limited scale. For example, MIDV-2020 [4] contains only 1,000 distinct template images across ten document categories. **(Category 2) Training-based approaches** [5], including those using GANs [6], differential privacy [7], generative AI [8], and diffusion models [9], can still require large numbers of labeled real documents that are difficult to obtain due to regulatory constraints, and hard to generalize to different countries and regions. **(Category 3) Few-shot approaches** [10], [11], [12] aim to minimize data requirements by using inpainting techniques to fill in synthetic personal data and face images into a template. Our prior work, IDNet [12], a representative of this category, has been downloaded more than **12,000** times (aggregated from eight parts) on the Zenodo platform since 2024, and used in diverse scenarios [13], [14], [15], underscoring both the unmet demand and the practical impact of generating large-scale synthetic documents with only one or a few real-world document examples. However, despite the cost-effectiveness of the few-shot approaches, they often produce content that diverges from the target domain distribution. This domain shift can mislead fraud detection models and result in unreliable evaluation outcomes.

Furthermore, existing methods lack flexible, declarative specificity over what documents to generate, yet comprehensive benchmarking requires testing under a wide range of conditions, such as documents from specific demographic groups, or scanned documents at particular rotation angles.

This paper introduces IDSPACE, a model-guided few-shot framework for generating high-quality synthetic identity documents with a flexible declarative task-specification interface,

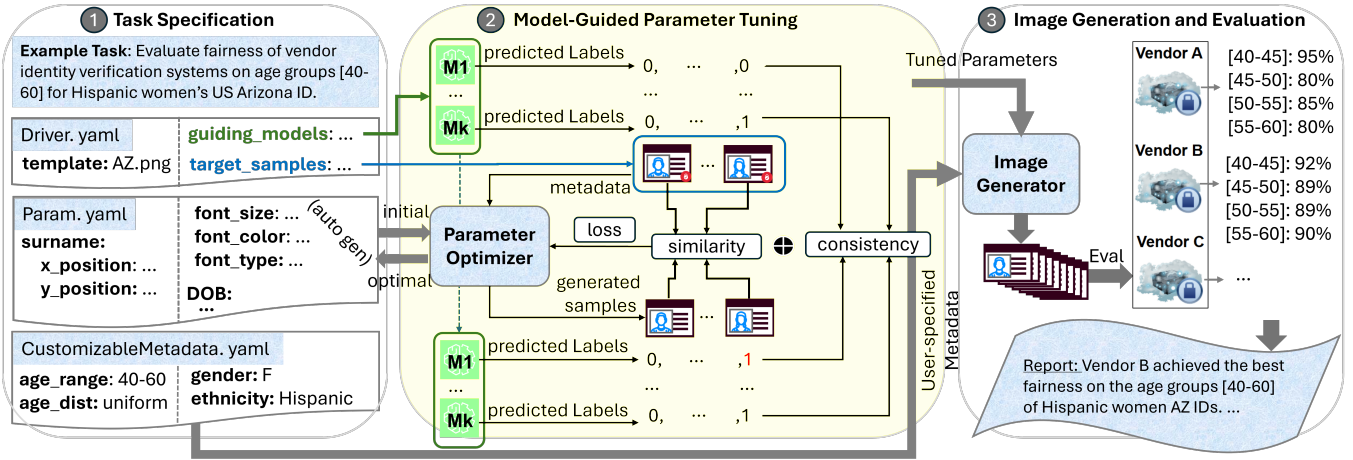


Fig. 1: Overview of the IDSPACE framework. The pipeline consists of three stages: (1) Task specification, where users specify high-level metadata constraints and provide a few real samples and models from the target domain; (2) Model-guided parameter tuning, which optimizes rendering parameters to maximize structural similarity and prediction consistency; and (3) Image generation and evaluation, which generates a synthetic benchmark dataset to evaluate remote identity verification systems and report results.

while enabling reliable and reproducible evaluation and training of identity verification systems, as illustrated in Fig. 1. First, IDSPACE introduces model-guided Bayesian optimization, which automatically tunes the generation parameters that control the filling of *user-specified metadata*, such as age, gender, ethnicity, and fraud patterns, into a user-specified template image as illustrated in Fig. 2 to maximize both visual similarity and prediction consistency with the target domain, using only a few real samples. Second, it decouples *user-specified metadata* from *automatically tuned control parameters* that govern low-level rendering details, such as font styles, font sizes, font colors, noise levels, brightness, and sharpness. This separation allows users to configure evaluations without low-level expertise. Third, IDSPACE expands beyond template images to support scanned and mobile-captured documents, enabling more realistic and diverse benchmarking settings.

Onfido’s survey of recent ID frauds [1] showed that an attacker can produce a convincing forged *physical* document, printed or digitally composited, and then capture it through a legitimate app on a real device. The resulting image has clean metadata and passes pipeline integrity checks, yet the underlying document may be fraudulent. IDSPACE is designed to support evaluation under such document-level forgery scenarios: detecting manipulated text fields and photos, inconsistent fonts, inconsistent fields, and other artifacts that distinguish forged documents from genuine ones, regardless of how the image was captured.

Concretely, we make the following contributions:

- **Reliable evaluation of identity verification systems under data scarcity.** IDSPACE reframes synthetic identity document generation from a data augmentation problem to a *benchmarking and evaluation* problem under severe data scarcity. While it builds on IDNet [12], which focuses on visual realism for training, IDSPACE is designed to ensure *evaluation validity* by preserving model behavior between real and synthetic documents. Rather than optimizing solely

for perceptual similarity, IDSPACE introduces a *prediction consistency* metric and *user-provided target models* that are trained on the target domain of documents, and it uses model-guided Bayesian optimization to auto-tune rendering (or “control”) parameters that preserve prediction consistency between the documents in the target domain and the generated synthetic documents. Therefore, it can reliably serve as a proxy for real data in downstream evaluation, a capability not supported by any prior synthetic datasets.

- **Specification-driven generation with decoupled user intent and rendering control.** Different from existing document generation frameworks, IDSPACE separates user-specified domain parameters (or “metadata”; the *what* to evaluate) from automatically tuned rendering parameters (the *how* to generate). Users specify metadata declaratively via YAML/JSON/CSV, while rendering parameters are optimized internally to support reproducible and controlled benchmarking. Tab. I formalizes this separation across template, scanned, and mobile formats.
- **A large-scale, multi-modal dataset for benchmarking identity verification systems.** We release a dataset of 359,240 synthetic documents across ten European ID types, spanning templated, scanned, and mobile-captured modalities, publicly available under CC-BY 4.0¹ with source code available under Apache 2.0 license². Experiments show that with as few as two real samples, IDSPACE improves evaluation consistency by 15–45% over CycleGAN [16], diffusion inpainting [17], and non-guided optimization [12], enabling reliable evaluation and benchmarks. The evaluation results also showed that the documents generated by IDSPACE have consistently better SSIM similarity with the documents in the target domain than IDNet. In addition, the accuracy of the models trained on IDSPACE outperformed IDNet by up to 9%.

¹<https://huggingface.co/datasets/cactuslab/IDSpace>

²<https://github.com/asu-cactus/IDSpace>



Fig. 2: Template of Albania ID card.

II. RELATED WORK

Public Synthetic Datasets and Synthetic Data Generation Methods for Identity Documents.

As mentioned, manually crafted datasets such as MIDV-500 [3], MIDV-2020 [4], MIDV-UP [18], and KID34K [19] suffer from the document generation cost, limited to around 100 templated documents for one country. SIDTD [20] and FMIDV [21] rely on MIDV-2020 to provide fraud patterns using inpainting or crop-and-move techniques. Overall, these datasets lack diversity and flexibility in performing comprehensive and customizable model evaluation tasks. Other large-scale datasets suffer from different quality issues, which fall into the following categories.

(1) Redaction-based document generation. For example, **BID** [22] redacts sensitive information such as portrait photos from real-world documents, which reduces their utility for many portrait-based fraud detection applications, e.g., face morphing detection. **(2) Training-based generation** [5] Generative-Adversarial Network (GAN) is widely used for synthetic document generation. For example, StyleGAN2 [23] was used to generate ID images building on its strong capability in synthesizing highly realistic human faces. However, it struggled with alphanumeric characters, and it also needed thousands of real IDs for model training, which is impractical in our target scenarios where only a small number of documents from the target domain of models to be evaluated is available. DP-DocLDM [9] uses private documents to finetune a public Diffusion model with differential privacy guarantees to generate documents, such as images of resumes, news articles, memos, scientific reports, etc. However, they do not focus on identity documents. In addition, their finetuning process used RVL-CDIP and Tobacco3482 datasets. The former consists of 400,000 document images, while the latter consists of more than 3,000 documents. CycleGAN [24] has been used to adapt low-fidelity identity documents to the target domain. However, this approach usually requires a large number of labeled samples of the target domain, which is not practical in our target scenarios. Ben-David et al. [25] proposed impossibility theorems, indicating that even with small distributional divergence or a universally good classifier, domain adaptation can fail without labeled target data. **(3) Few-shot Approaches.** DocXPand-25k [11] applies inpainting to fill metadata into a self-designed document template, and it can hardly generalize

to real-world document types. Our prior work IDNet [26], [12] is a low-cost framework that tunes a couple of parameters to maximize the similarity between the generated documents and target domain documents. While it is instrumental in augmenting training data and its parts have been downloaded for more than 12,000 times in total on Zenodo, its quality issues have been noticed [18]. We further identified a prediction inconsistency using multiple fraud-detection models between the target domain and the IDSPACE-generated documents, due to an observed discrepancy between the generated dataset and the target domain, as illustrated in Fig. 3(f). In this work, we addressed this limitation by introducing a novel model-guided data generation technique. In addition, IDNet only involved templated document images (i.e., images with information filled in the fields of templates), while this work has expanded the modality to documents captured by scanners and mobile phones. Our experiments in Sec. VII showed that compared to IDNet, IDSPACE significantly improved the prediction consistency, training accuracy, and SSIM similarity with the documents from the target domain.

Synthetic Data Generation for Model Evaluation. [27] discusses adaptation of the synthetic data to the real domain using GANs. Van Breugel et al. [28] proposed to use a deep generative model on the test dataset to create synthetic data for evaluating model performance on underrepresented subgroups and under distributional shifts. However, their approach relies on the availability of test data, which is often scarce in practice in our targeting scenarios.

III. A NOVEL PROBLEM ABSTRACTION

In this section, we present a few-shot synthetic data generation methodology designed to balance cost and quality in evaluating identity verification systems. Our approach is to combine the decoupling of the user-specified metadata and the control parameters finetuned by a model-guiding framework.

Tab. I illustrates how we distinguish metadata that should be explicitly specified by users from parameters that should be automatically adjusted in this work. Parameters that can be reliably inferred using external tools are excluded. For instance, the background color of a portrait photo can often be extracted using standard color analysis tools and does not require automated tuning. In contrast, detecting text color when it overlaps with complex background images is significantly more difficult, and identifying the original font styles used in official identity documents is often infeasible due to their proprietary nature. In such cases, we rely on auto-tuning. Importantly, advanced users can redefine the boundary between user-specified metadata and automatically controlled parameters, depending on their available tools, expertise, and resources. Based on this flexible and modular design, we are the first to formalize the synthetic identity document generation problem as follows.

Portrait Photos. By default, portrait photos are selected from an open academic dataset published by generated.photos [29] that consists of 10,000 portrait photos, with 5,979 photos suitable for identity documents. For each document, the

Customized Template Documents



Fig. 3: Examples and t-SNE comparison for non-fraud documents generated by IDSPACE, IDNet [12], and MIDV-2020 (the target domain).

TABLE I: Overview of Exemplar Synthetic ID Generation Parameters

	Control Parameters (AutoTuned)	User-Customizable Metadata
Customized Template Documents	Font: Font style and size of each text field. Text color: Color of texts that overlap with the background. Positions: Positions of each field in the template, and character spacing within each field. Quality: JPEG image quality settings (e.g., quality factor in [50–95]).	ID Template: File path of the template image used for ID generation. Fraud Pattern: Whether to simulate fraud and which type—e.g., <i>crop-and-move</i> , <i>inpaint-and-rewrite</i> , following SIDTD [20]. Entity Information: First name, last name, ID number, DOB, ethnicity group, gender, height, weight, issue date, expiration date, etc.
Scanned/Mobile Documents	Noise Level: Amount of Gaussian noise added—e.g., standard deviation in range [0, 25]. Subtle Blurring: Gaussian blur with sigma in range [0.5, 2.0] to simulate out-of-focus scans. Brightness: Adjustment factor for image brightness—e.g., randomly sampled from [0.8, 1.2]. Contrast: Adjustment factor for image contrast—e.g., randomly sampled from [0.8, 1.2]. Sharpness: Strength of sharpening filter—e.g., factor in [0.5, 2.0].	ID Template Image: File path of the ID image (a customized template document) to be scanned. Resolution: DPI value, e.g., 200, 300, or 400. Color Mode: One of {"color", "grayscale", "black and white"}. Position/Orientation (for scanned images): Placement of the ID—e.g., rotated by $\pm 5^\circ$ or shifted ± 10 px. Background image (for mobile images): An image to serve as the background of the ID placement.



Fig. 4: Example Mobile document images generated.

selection of the portrait photo must be consistent with the specified metadata, which means the user-specified age (DOB), gender, ethnicity group, and weight, must match the face of the portrait photo. This is not difficult given that each photo in the generated.photos dataset has all annotated information specifying age, gender, and ethnicity groups. Advanced users can also replace the generated.photos dataset with their own portrait photo database.

Fraud Patterns. As shown in Tab. I, users can specify the numbers of non-fraud and fraud documents, as well as the distribution of fraud patterns among the fraud documents.

Our current implementation supports two representative fraud patterns identified in Onfido/Entrust’s *Identity Fraud Report 2024* [1]: (1) *Inpaint-and-Rewrite*. A text field is randomly selected from all available ID fields, excluding the portrait photo and signature. A realistic mask is applied to the selected field region, and the replacement text is rendered using a randomly selected font size and style. (2) *Crop-and-Move*. A field, such as last name, date of birth, or address, is cropped from one ID and replaced with the corresponding field from another ID. In both patterns, fields are selected randomly, with a 95% probability of selecting the same field across the two IDs and a 5% probability of selecting different PII fields, following the standard implementations of SIDTD [20] and IDNet [12]. These are also the only two fraud patterns supported by both SIDTD and IDNet. (IDNet supports additional fraud patterns: face morphing, photo replacement, and a trivial mixture mode. Since these patterns are confined to the portrait photo field and do not affect other ID fields, incorporating them into our framework is straightforward. For example, users can provide a set of portrait photos with morphed faces in the metadata to replace the default generated.photos dataset [29]. We exclude these photo-based fraud patterns from our evaluation because mature benchmarks already exist for face morphing detection, such as FRLL, FERET, and FRGC, and inserting morphed or

replaced portraits into synthetic documents does not introduce additional document-level generation challenges.)

Problem Definition. Given a template of a type of ID document (e.g., a template of the Albania ID card, as illustrated in Fig. 2), denoted as T , generating the i -th ID image has two steps: (1) obtaining or generating the metadata information x_{meta}^i following user-specification, which includes fraud patterns, capturing device (e.g., scanner), capturing environments (e.g., rotation and position of the document, color mode, and resolution), and various personal information (e.g., first name, last name, date of birth, ID card number, portrait photo, eye color, height, weight, card issue date, expiration date), listed as user-customizable metadata in Tab. I. (2) filling in the metadata information into the template to generate the final image x_i , denoted as $x_i = G_\theta(x_{meta}^i, T)$. Here, θ represents the parameters that control filling the metadata into the template, such as those control parameters listed in Tab. I. While $G_\theta(\cdot)$ represents the process of transforming the metadata x_{meta}^i and the given template T into a synthetic ID image x_i using the control parameters θ .

Given an existing machine learning model f trained for fraud detection for a target domain consisting of ID documents sharing the same template T , denoted as $\mathcal{D}_{real} = \{x_i\}$, with f 's training dataset $\mathcal{D}_{training} \subset \mathcal{D}_{real}$. Assuming each sample $x_i \in \mathcal{D}_{real}$ having metadata $meta(x_i)$, given a small number of samples from $\mathcal{D}_{real}$, we would like to learn θ so that $\forall x_i \in \mathcal{D}_{real}$, we have $x_i = G_\theta(meta(x_i), T)$, and thus $f(x_i) = f(G_\theta(meta(x_i), T))$.

IV. MODEL GUIDED CONTROL PARAMETER TUNING

To address the problem described in Sec. III, we propose combining Bayesian optimization [30] with our custom optimization objective that introduces l guiding models $f_1, \dots, f_l$, which are trained in the target domain. *Importantly, these k models are not necessarily the target models to be evaluated.* The objective is not only to maximize the overall similarity (e.g., measured using structural similarity index measure (SSIM)) between each input document x_i and the corresponding generated document $G_\theta(meta(x_i), T)$ for $i = 1, \dots, m$ (See Eq. 2), but also to improve evaluation consistency between $f_k(x_i)$ and $f_k(G_\theta(meta(x_i), T))$ for $i = 1, \dots, m$ and $k = 1, \dots, l$ (See Eq. 3). Here, m denotes the number of samples from the target domain, which is assumed to be small, given the scarcity of ID data. Our custom optimization objective is formalized in Eq. 1, which is a weighted sum of the similarity metric (Eq. 2) and the evaluation consistency metric (Eq. 3).

$$\theta^* = \arg \max_{\theta \in \Theta} (\lambda_0 \cdot \mu_{similarity}(\theta) + \dots + \lambda_l \cdot \mu_{consistency}^l(\theta)) \quad (1)$$

$$\mu_{similarity}(\theta) = \frac{1}{m} \sum_{x_i \in \mathcal{D}_{real}} similarity(x_i, G_\theta(meta(x_i), T)) \quad (2)$$

$$\mu_{consistency}^k(\theta) = \frac{1}{m} \sum_{x_i \in \mathcal{D}_{real}} \mathbb{1}(f_k(x_i) = f_k(G_\theta(meta(x_i), T))) \quad (3)$$

Our proposed Bayesian optimization algorithm formalized in Alg. 1 first trains a surrogate model to learn the relationship

Algorithm 1 Model-Guided Bayesian Optimization of Control Parameters (For simplicity, it only used one guiding model f , and the extension to multiple guiding models as illustrated in Eq. 1 is trivial)

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1: Input: A small dataset from  $\mathcal{D}_{real}$ :  $\mathcal{D}_{sample} = \{x_1, x_2, \dots, x_m\}$ , a template  $T$ 
2: synthetic data generator  $G$ , the guiding model  $f$ , parameter space  $\Theta$ , weight  $\lambda_0, \lambda_1$ 
3: Output: Optimized parameter set  $\theta^*$ 
4: Initialize Bayesian Optimization over  $\Theta$  ▷ Setup BO framework
5: for iteration  $t = 1$  to  $T$  do
6:    $\theta_t \leftarrow$  BO acquisition function ▷ Sample candidate parameters
7:    $\mathcal{D}_{syn} \leftarrow \{G_{\theta_t}(meta(x_i), T) | x_i \in \mathcal{D}_{sample}\}$  ▷ Generate synthetic data
8:    $\mu_{similarity}^t \leftarrow \frac{1}{m} \sum_{i=1}^m similarity(x_i, \mathcal{D}_{syn}^{(i)})$  ▷ Compute similarity, e.g. SSIM (Eq. 2)
9:    $\mathbf{p}_{real} \leftarrow f(\mathcal{D}_{sample})$  ▷ Get model prediction results
10:   $\mathbf{p}_{syn} \leftarrow f(\mathcal{D}_{syn})$ 
11:   $\mu_{consistency}^t \leftarrow \frac{1}{m} \sum_{i=1}^m \mathbb{1}(p_{real}^{(i)} = p_{syn}^{(i)})$  ▷ Compute consistency (Eq. 3)
12:   $J(\theta_t) \leftarrow \lambda_0 \mu_{similarity}^t + \lambda_1 \mu_{consistency}^t$  ▷ Compute Objective score (Eq. 1)
13:  Update BO model with  $(\theta_t, J(\theta_t))$ 
14: end for
15: return  $\theta^* = \arg \max_{\theta \in \{\theta_1, \dots, \theta_T\}} J(\theta)$  ▷ Select best parameters

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Algorithm 2 Synthetic Data Generation

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1: Input: User-specified metadata:  $\theta_{user}$ ; control parameters tuned using Alg. 1:  $\theta^*$ ; ID template:  $T$ ; Synthetic data generator:  $G$ ; Metadata generator:  $F$ 
2: Output: Generated document  $x_{syn}$ 
3:  $meta \leftarrow F_{\theta_{user}}()$  ▷ Generate metadata based on user specification
4: return  $x_{syn} = G_{\theta^*}(meta, T)$  ▷ Generate data using tuned control parameters

```

between θ (the control parameters of the data generation process) and an objective metric as formalized in Eq. 1. It then iteratively selects θ guided by the surrogate model to optimize the objective. In each iteration, it evaluates the effectiveness of the selected θ using the objective function, and use the measured results to update the surrogate model. The Bayesian optimization strategy can be replaced with other search methods, such as Hyperband [31], which are evaluated and compared to our approach in Sec. VII-C.

V. DOCUMENT GENERATION

After optimizing the control parameters using Alg. 1, we apply a synthetic data generation process formalized in Alg. 2. It consists of two steps:

Step 1. Generating metadata following user specification (line 3 in Alg. 2). By customizing the metadata, users can control which aspects of the model performance to focus on during the evaluation process. For example, if a user needs to evaluate the model's fraud detection fairness on different gender, age, or ethnicity groups, s/he may request to generate documents with uniform distributions across all groups, including minority groups. Users can specify metadata by either uploading a CSV file that lists metadata for each document or a YAML file that defines the probability distribution of each type of metadata, used as θ_{user} in Alg. 2.

Step 2. Generating the documents (line 4 in Alg. 2). Customized template documents, as shown in Fig. 1 and Fig. 3, emphasize geometric precision in text alignment and compliance with governmental design specifications. Such quality requirements are satisfied via the control parameters (θ^*) tuned by our model-guided Bayesian optimization algorithm and the function ($G_{\theta^*}(\cdot)$) that applies those parameters to drive

varying-scale synthesis by filling in the template following user-specified information. In addition to the generated document, each output includes annotations such as typographic specifications, positional coordinates, semantic segment masks, and bounding boxes in JSON format, facilitating downstream tasks, e.g., constructing scanned or mobile documents.

Taking scanned documents as an example, the pipeline first loads the user-specified customized template document. Then, it applies the tuned control parameters to fill user-specified metadata into the template image. After that, the function generates the scanner background image and combines the transformed customized template document and the background following the user-specified position and rotation parameters.

The generation of mobile documents is similar. A user can flexibly specify to use the background of a certain existing mobile ID image (e.g., a picture of A’s driver’s license (DL) placed on top of the keyboard of a computer), and a template image, e.g., B’s DL, to be blended by replacing A’s DL in the image with B’s DL. We used the MIDV’s collection of mobile documents, users can specify to generate a new mobile dataset with $x\%$ of indoor backgrounds and $1 - x\%$ of outdoor backgrounds. Users can also specify the distribution of age, gender, ethnicity groups, and fraud patterns of the entities involved in the new template images to be blended with existing documents. To generate the mobile and scan documents, we employed a combination of advanced computer vision models and image processing techniques. In particular, we use Grounding DINO [32], a state-of-the-art model that integrates object detection and language grounding, to detect and localize the document in the original background image. Once localized, the Segment Anything Model (SAM) [33] is applied to obtain an accurate segmentation mask of the original document. The prompt-based interface of SAM allows for precise and flexible segmentation, which is critical for accurate geometric alignment.

For the blending stage, we adopt the Deep Image Blending (DIB) framework [34], which synthesizes high-quality images by optimizing a combination of loss functions including Poisson gradient loss, content loss, style loss, histogram loss, and total variation loss. To enhance structural fidelity, we extend the DIB loss with an additional differentiable Structural Similarity Index (SSIM) loss. This augmentation improves both local and global consistency between the blended image and the background, as detailed in our extended technical report [35].

VI. IDSPACE DATASET

We first extracted 5,979 adult entities from the generated.photos [29] dataset. Each entity has metadata information, such as gender, ethnicity group, and age, and a photo. Among these entities, we have 3,009 females, and 2,970 males. Among females, there are 825 entities associated with the Asian ethnicity group, 925 with Latino, 467 with black, and 752 with white. Among the males, there are 612 with Asian, 787 with Latino, 746 with black, and 825 with white. We used this information as user-specified metadata, of which the distribution is consistent with the IDNet[36] dataset.

We used ten templates of identity document types from ten European countries, from our prior IDNet work, including the ID cards and passports Albania, Azerbaijan, Estonia, Finland, Greece, Latvia, Russia, Serbia, Slovakia, and Spain.

For each of these 5,979 entities, using our IDSPACE data generation methodology detailed in Sec. III to Sec. V, we generated one non-fraud document, two fraud documents with the *crop-and-move* fraud pattern and the *inpaint-and-replacement* fraud pattern, respectively, for each of the ten European country identity document types. In the process, we used the ResNet50 model detailed in Sec. VII-A1 as the guiding models. We used the *tuning set* with 40 documents with balanced labels from the target domain to tune the control parameters.

Then, similarly, for each of these generated documents, we further create one scanned document using randomly selected positions and rotations. We also generated 50 mobile documents for each of the ten European identity document types. For each mobile document, we randomly sample one mobile ID document from the MIDV dataset as the background, and one templated document we generated for the given document type.

In total, our new dataset has 359,240 images, including 179,370 templated documents, 179,370 scanned documents, and a small set of 500 mobile documents. We published our dataset on HuggingFace (See footnote 1) for public access.

VII. EVALUATION OF IDSPACE DATA GENERATOR

We conducted a comprehensive empirical study to investigate the following research questions. R1. Will models trained on the target domain achieve consistent evaluation results on our generated document and the corresponding document from the target domain with the same metadata, and will the evaluation (prediction) consistency of our generated documents outperform alternatives? R2. Does the dataset generated by IDSPACE also benefit the learning process? R3. How does our proposed model-guided parameter tuning approach compare to alternatives?

System Environment. All experiments were conducted on an Ubuntu Linux server equipped with 48 CPU cores (Intel Xeon Silver 4310, 2.10 GHz), 125 GB of RAM, and two NVIDIA A10 GPUs (24GB VRAM each). The server is installed with 256GB NVMe SSD and 1TB HDD drive.

Experiment Setup. We used the MIDV [37] and SIDTD [38] datasets, which are under the CC BY-SA 2.5 and CC-BY-4.0 licenses, as the target domain. They include ten types of European country identity documents, with 100 templated documents and around 100 fraud templated documents in each type. We sampled 50% of documents to form a *training set* for learning the fraud detection models, which are used as the guiding models for parameter tuning and the target models to be evaluated. 30% of samples form the *testing set* to evaluate the consistency of the prediction of the model of all the baseline data generation methods. The additional 20% of the documents compose a *tuning set* to adjust the control parameters following Alg. 1. Due to space limitations, our evaluation focuses on customized template images, which is fundamental to identity fraud detection [20], [37], [26]. We put the corresponding

evaluation results for scanned documents in our extended technical report [35]. In our implementation of Alg. 1, we used SSIM as our similarity metric. We set λ_0 to 1 and λ_i to $1/l$ (l is the number of guiding models, $i = 1, \dots, l$) to balance similarity and consistency. The tuning of λ_i is discussed in the technical report [35].

A. R1. Prediction Consistency Comparison

1) *Using Models Independently Finetuned on the Target Domain:* To answer R1, we first selected five fraud detection models with model architectures, including ViT-large, ResNet50, Inception-v3, VGG16, and DenseNet, and finetuned them independently using documents from the target domain, achieving accuracies of 1.0000, 1.0000, 1.0000, 0.9873, and 1.0000, respectively. We selected those models because they are widely adopted in academia and industry for fraud detection on identity documents [20], [19], [1], [39], [40], [41], [42], [43]. We then used each of the following baselines with the *tuning set* to generate a dataset, using the same metadata as the samples from the *testing set*, which also came from the target domain. We further compared the model prediction consistency, defined in Eq. 3, between the dataset generated by each baseline and the *testing set* for each of the five models.

- **BO w/ SSIM-only objective:** This approach represents IDNet, which did not adopt any model guidance, i.e., using Eq. 2 as the objective function of the Bayesian Optimization (BO) search.

- **CycleGAN:** This is a widely used domain adaptation approach with pre-trained models publicly available [16], [44] under the BSD license. We chose the CycleGAN model pretrained on the Flickr dataset [45], finetuned it using the *tuning set*, and used the finetuned model to adapt an ID dataset generated by vanilla BO (i.e., BO w/ SSIM-only objective) to the target domain.

- **Diffusion-based inpainting:** We include a diffusion-based model as a baseline using image-to-image generation with inpainting. Specifically, we use the pretrained Stable Diffusion v1.5 inpainting model (stable-diffusion-v1-5/stable-diffusion-inpainting) [17]. Given an input identity document image, a text prompt specifying the target segment, and a mask indicating the editable region, the model generates an output image in which the segment is updated while the remaining content is preserved. This baseline enables comparison between diffusion-based inpainting and other approaches for controlled identity document editing.

- **IDSPACE:** This is our approach using different combinations of guiding models for BO search.

Prediction Consistency The comparison results are presented in Tab. II. We observed significant improvement of consistency scores using our proposed model-guided Bayesian optimization methodology, ranging from 15.62% to 30.55% compared to BO w/ SSIM-only objective, 27.40% to 42.33% compared to CycleGAN, and 30.60% to 45.51% compared to Diffusion-based inpainting. We also found that using our proposed approach, the consistency is not only improved for the guiding models used in the BO objective function but also for other models

trained on the target domain. In addition, incorporating more guiding models further improved consistency in the majority of the cases. The algorithm demonstrated robustness across different architectures (mean consistency = $0.9354 \pm 0.003\%$). Using our approach, small models such as Inception-v3 and DenseNet are easier to achieve better consistency than other larger models.

To investigate how the model prediction consistency changes with the number of samples available from the target domain for different baselines, we applied 2, 20, and 40 samples (with balanced fraud and non-fraud labels) from the *tuning set* to finetune the CycleGAN model, and to tune the control parameters for the baseline using BO w/ SSIM-only objective and our model-guided IDSPACE approach. We used the *testing set* that is disjoint with the *tuning set* to measure the prediction consistency of the ResNet50 model used in Tab. II between the documents from the *testing set* and the generated documents, both of which share the same metadata (i.e., field values and photos). Our IDSPACE approach used the target ResNet50 model as the guiding model.

The results in Tab. III showed that our approach consistently and significantly improved prediction consistency even when fewer samples are available from the target domain. This result demonstrated the benefits of incorporating guiding models into the search objective for a few-shot approach. Furthermore, the overall objective function improved as the number of samples from the target domain increased. Notably, even with just two samples, our algorithm demonstrated substantial improvement compared to baselines, proving the effectiveness of our proposed model-guided framework.

Fidelity. We further evaluate the SSIM similarity between MIDV documents and the documents generated by IDNet and IDSPACE across multiple target regions over multiple test samples, as shown in Fig. 5. IDSPACE consistently outperforms IDNet across all regions. Notably, substantial gains are achieved even with a single guidance sample, and increasing the number of guidance samples from 1 to 10 or 20 does not consistently improve SSIM and can slightly degrade performance in some regions. This behavior suggests that a small number of guidance samples induces a strong, coherent structural prior that generalizes well across the test set, whereas incorporating additional guidance samples increases structural variability, leading to a more flexible but less tightly constrained mapping. Since SSIM is particularly sensitive to fine-grained structural alignment, this increased variability can lead to slightly lower average SSIM scores, while still maintaining clear improvements over IDNet.

Control Parameter Value Variation during the Model-Guided Optimization Process. Models optimized jointly usually (e.g., two- or three-model combinations) exhibit less variance than their single-model counterparts. This indicates that multi-model BO acts as a regularizer: because the joint objective must satisfy multiple architectures simultaneously, the search converges toward parameter regions that are robust across models, resulting in more stable and predictable behavior.

TABLE II: Model prediction consistency evaluation on different models with best results highlighted in blue

	Model Prediction Consistency with Target Test Data (Mean $\pm$ Std)					Average(row)
	ViT-Large	ResNet50	Inception-v3	VGG16	DenseNet	
CycleGAN	0.5648 $\pm$ 0.004	0.5324 $\pm$ 0.000	0.5093 $\pm$ 0.000	0.5648 $\pm$ 0.000	0.5000 $\pm$ 0.000	0.5343 $\pm$ 0.001
Diffusion-based inpainting	0.5000 $\pm$ 0.000	0.4958 $\pm$ 0.000	0.5000 $\pm$ 0.000	0.5167 $\pm$ 0.000	0.5000 $\pm$ 0.000	0.5025 $\pm$ 0.000
BO w/ SSIM-only objective (IDNet)	0.7257 $\pm$ 0.018	0.5382 $\pm$ 0.006	0.8646 $\pm$ 0.012	0.6146 $\pm$ 0.006	0.5174 $\pm$ 0.012	0.6521 $\pm$ 0.017
IDSPACE						
Guiding models						
DenseNet	0.8819 $\pm$ 0.093	0.8646 $\pm$ 0.018	0.9965 $\pm$ 0.006	0.8715 $\pm$ 0.040	1.0000 $\pm$ 0.000	0.9229 $\pm$ 0.005
DenseNet + Inception-v3	0.8021 $\pm$ 0.136	0.8750 $\pm$ 0.017	0.9931 $\pm$ 0.007	0.8715 $\pm$ 0.060	1.0000 $\pm$ 0.000	0.9083 $\pm$ 0.007
DenseNet + Inception-v3 + ResNet50	0.8194 $\pm$ 0.091	0.9306 $\pm$ 0.017	1.0000 $\pm$ 0.000	0.8889 $\pm$ 0.028	1.0000 $\pm$ 0.000	0.9278 $\pm$ 0.006
DenseNet + Inception-v3 + VGG16	0.8924 $\pm$ 0.015	0.8646 $\pm$ 0.036	1.0000 $\pm$ 0.000	0.9410 $\pm$ 0.025	1.0000 $\pm$ 0.000	0.9396 $\pm$ 0.004
DenseNet + Inception-v3 + ViT-Large	0.9306 $\pm$ 0.020	0.8889 $\pm$ 0.045	0.9965 $\pm$ 0.006	0.9097 $\pm$ 0.036	1.0000 $\pm$ 0.000	0.9451 $\pm$ 0.003
DenseNet + ResNet50	0.9340 $\pm$ 0.015	0.9236 $\pm$ 0.016	1.0000 $\pm$ 0.000	0.8819 $\pm$ 0.035	1.0000 $\pm$ 0.000	0.9479 $\pm$ 0.003
DenseNet + ResNet50 + VGG16	0.9236 $\pm$ 0.029	0.8924 $\pm$ 0.006	1.0000 $\pm$ 0.000	0.9375 $\pm$ 0.016	1.0000 $\pm$ 0.000	0.9507 $\pm$ 0.002
DenseNet + ResNet50 + ViT-Large	0.9201 $\pm$ 0.033	0.9062 $\pm$ 0.041	0.9792 $\pm$ 0.036	0.8403 $\pm$ 0.094	0.9965 $\pm$ 0.006	0.9285 $\pm$ 0.004
DenseNet + VGG16	0.9132 $\pm$ 0.012	0.8819 $\pm$ 0.029	1.0000 $\pm$ 0.000	0.9306 $\pm$ 0.034	1.0000 $\pm$ 0.000	0.9451 $\pm$ 0.003
DenseNet + VGG16 + ViT-Large	0.9306 $\pm$ 0.017	0.8542 $\pm$ 0.030	1.0000 $\pm$ 0.000	0.9479 $\pm$ 0.032	1.0000 $\pm$ 0.000	0.9465 $\pm$ 0.004
DenseNet + ViT-Large	0.9306 $\pm$ 0.039	0.8646 $\pm$ 0.060	1.0000 $\pm$ 0.000	0.8958 $\pm$ 0.029	1.0000 $\pm$ 0.000	0.9382 $\pm$ 0.004
Inception-v3	0.6146 $\pm$ 0.108	0.7778 $\pm$ 0.057	0.9931 $\pm$ 0.007	0.7118 $\pm$ 0.129	0.9444 $\pm$ 0.056	0.8083 $\pm$ 0.025
Inception-v3 + ResNet50	0.7708 $\pm$ 0.068	0.9340 $\pm$ 0.030	0.9965 $\pm$ 0.006	0.8681 $\pm$ 0.012	0.9965 $\pm$ 0.006	0.9132 $\pm$ 0.009
Inception-v3 + ResNet50 + VGG16	0.9375 $\pm$ 0.023	0.9097 $\pm$ 0.025	0.9965 $\pm$ 0.006	0.9479 $\pm$ 0.027	0.9965 $\pm$ 0.006	0.9576 $\pm$ 0.001
Inception-v3 + ResNet50 + ViT-Large	0.9375 $\pm$ 0.007	0.9271 $\pm$ 0.027	0.9965 $\pm$ 0.006	0.8785 $\pm$ 0.027	1.0000 $\pm$ 0.000	0.9479 $\pm$ 0.003
Inception-v3 + VGG16	0.9375 $\pm$ 0.030	0.8750 $\pm$ 0.026	1.0000 $\pm$ 0.000	0.9549 $\pm$ 0.025	1.0000 $\pm$ 0.000	0.9535 $\pm$ 0.003
Inception-v3 + VGG16 + ViT-Large	0.9410 $\pm$ 0.030	0.8194 $\pm$ 0.033	1.0000 $\pm$ 0.000	0.9410 $\pm$ 0.025	1.0000 $\pm$ 0.000	0.9403 $\pm$ 0.005
Inception-v3 + ViT-Large	0.9340 $\pm$ 0.023	0.8576 $\pm$ 0.021	0.9965 $\pm$ 0.006	0.9062 $\pm$ 0.032	1.0000 $\pm$ 0.000	0.9389 $\pm$ 0.004
ResNet50	0.9271 $\pm$ 0.015	0.9514 $\pm$ 0.012	1.0000 $\pm$ 0.000	0.8993 $\pm$ 0.023	1.0000 $\pm$ 0.000	0.9556 $\pm$ 0.002
ResNet50 + VGG16	0.9097 $\pm$ 0.016	0.9097 $\pm$ 0.029	1.0000 $\pm$ 0.000	0.9306 $\pm$ 0.014	1.0000 $\pm$ 0.000	0.9500 $\pm$ 0.002
ResNet50 + VGG16 + ViT-Large	0.9375 $\pm$ 0.012	0.9167 $\pm$ 0.026	1.0000 $\pm$ 0.000	0.9132 $\pm$ 0.006	1.0000 $\pm$ 0.000	0.9535 $\pm$ 0.002
ResNet50 + ViT-Large	0.9132 $\pm$ 0.023	0.9340 $\pm$ 0.050	0.9965 $\pm$ 0.006	0.8785 $\pm$ 0.032	1.0000 $\pm$ 0.000	0.9444 $\pm$ 0.003
VGG16	0.9271 $\pm$ 0.040	0.8368 $\pm$ 0.054	0.9965 $\pm$ 0.006	0.9306 $\pm$ 0.052	1.0000 $\pm$ 0.000	0.9382 $\pm$ 0.004
VGG16 + ViT-Large	0.9514 $\pm$ 0.021	0.8542 $\pm$ 0.025	1.0000 $\pm$ 0.000	0.9444 $\pm$ 0.017	1.0000 $\pm$ 0.000	0.9500 $\pm$ 0.004
ViT-Large	0.9375 $\pm$ 0.016	0.8646 $\pm$ 0.065	0.9861 $\pm$ 0.010	0.8750 $\pm$ 0.024	0.9965 $\pm$ 0.006	0.9319 $\pm$ 0.004
Average(Column)	0.8982 $\pm$ 0.006	0.8846 $\pm$ 0.002	0.9969 $\pm$ 0.000	0.8999 $\pm$ 0.003	0.9972 $\pm$ 0.000	

TABLE III: Comparison of consistency using different numbers of target samples (the first column) with the best results in blue.

	CycleGAN	Diffusion	IDNet	IDSPACE
2	0.546 $\pm$ 0.000	0.500 $\pm$ 0.000	0.413 $\pm$ 0.000	0.806 $\pm$ 0.008
20	0.542 $\pm$ 0.000	0.500 $\pm$ 0.000	0.170 $\pm$ 0.000	0.944 $\pm$ 0.000
40	0.532 $\pm$ 0.000	0.496 $\pm$ 0.000	0.538 $\pm$ 0.006	0.951 $\pm$ 0.012

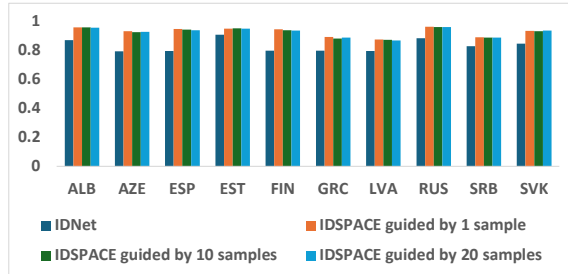


Fig. 5: Structural Similarity (SSIM) comparison between IDNet and MIDV vs. IDSPACE and MIDV, guided by different numbers of target-domain samples across multiple regions.

For most of the models, parameters such as x_c , y_c , and z_c , which encode RGB color channels, $font_style$ and $font_size$, and $stroke_w$, representing stroke width, exhibit high variance, indicating their significant impact on the effectiveness of generated examples.

2) *Limitations of Generative AI Models:* We evaluated representative generative approaches—including StyleGAN [23], diffusion-based image generation [17] (i.e., text-to-image,

different from the Diffusion-based inpainting), and recent large generative models such as GPT-4o and GPT-image-1 from OpenAI [46], under few-shot and limited-data regimes consistent with real-world identity verification settings. StyleGAN and diffusion models trained with limited target-domain data (e.g., up to 40 images to be consistent with our few-shot based approach) fail to preserve the fine-grained structural properties of identity documents, exhibiting distortions in layout, typography, and field alignment, due to the data hungry nature of these models. While large generative models, such as GPT-4o and GPT-image-1, demonstrate stronger global coherence in few-shot prompting scenarios, they similarly struggle to maintain precise document structure and layout fidelity required for reliable benchmarking. Results including output image examples can be found in our extended technical report [35]. As a result, documents generated by these approaches struggle to compete with our approach when working with a small number of samples from the target domain, limiting their suitability for evaluation under data scarcity.

3) *Can LLM detect IDSPACE documents as generated or synthetic?:* We conducted experiments to test whether existing LLM models such as GPT-4o could detect that our generated documents are generated. As shown in Tab. V, we tested using both zero-shot and few-shot settings. For the few-shot setting, we provided 2, 4, and 6 examples in the generated and real categories, respectively, but GPT-4o achieved only limited discrimination performance across all settings, indicating that the generated documents closely resemble real documents and are difficult for the model to reliably identify as synthetic.

	Target Models			
	ResNet50	Inception-v3	DenseNet	EfficientNet
IDNet	0.9156 $\pm$ 0.016	0.9536 $\pm$ 0.006	0.8734 $\pm$ 0.012	0.9873 $\pm$ 0.000
IDSPACE	0.9866 $\pm$ 0.000	0.9958 $\pm$ 0.000	0.9536 $\pm$ 0.006	0.9958 $\pm$ 0.000

TABLE IV: Utility (detection accuracy of the copy-and-move frauds) of the models trained on IDNet and IDSPACE, w/ best values highlighted in blue.

TABLE V: Results of using GPT-4o to detect whether our generated documents are real or generated.

Methods	ACC	PRC	Recall	F1
Zero-shot	0.4667 $\pm$ 0.058	0.4167 $\pm$ 0.144	0.1167 $\pm$ 0.020	0.1801 $\pm$ 0.04
Few-shot (2 demonstrations for real and generated respectively)	0.5167 $\pm$ 0.058	0.5170 $\pm$ 0.058	0.5167 $\pm$ 0.076	0.5159 $\pm$ 0.061
Few-shot (4 demonstrations for real and generated respectively)	0.5917 $\pm$ 0.076	0.5796 $\pm$ 0.060	0.6667 $\pm$ 0.126	0.6177 $\pm$ 0.081
Few-shot (6 demonstrations for real and generated respectively)	0.5750 $\pm$ 0.090	0.5693 $\pm$ 0.077	0.5833 $\pm$ 0.144	0.5747 $\pm$ 0.107

B. R2. Does the dataset generated by IDSPACE also benefit the learning process?

As demonstrated in Sec. VII-A, our proposed approach significantly improved the evaluation consistency of the generated documents with the *testing set*. Next, we will evaluate whether the fraud detection models trained on our generated dataset could also generalize well to the target domain.

To compare the utility of the IDSPACE-generated dataset described in Sec. VI to the IDNet[36] dataset, we trained different fraud detection models on each dataset and evaluated the utility of these models using the evaluation dataset from MIDV [4] and SIDTD [5] (i.e., the target domain). The results are shown in Tab. IV, which demonstrates the excellent utility of our IDSPACE framework in learning fraud detection tasks in the target domain, outperforming IDNet by up to 9%.

C. R3. Tuning Algorithm Comparison

In this section, we compare our model-guided Bayesian optimization (Bayesian) method with Hyperband, which accelerates the search for optimal configurations by adaptively allocating resources to promising candidates using early-stopping and successive halving [31]. For both approaches, we used 40 (i.e., 20%) samples from the *tuning set*, and ResNet50 served as both the guiding and target model. In our approach, hyperparameters *init_point* and *n_iter* control the accuracy vs latency tradeoff. Similarly, in Hyperband search, *max_resources* controls the maximum amount of resources that can be allocated to a single configuration, and η controls the proportion of configurations discarded in each round of successive halving. Let us denote each instance of $(init_point, n_iter)$ pairs and $(max_resources, \eta)$ pairs as b_i and h_i , respectively. In Fig. 6, the evaluated $\{b_1, b_2, \dots, b_6\}$ (green points) for our approach are $\{(50, 100), (50, 150), (50, 200), (100, 400), (100, 600), (100, 800)\}$, and $\{h_1, h_2, \dots, h_6\}$ (red points) for Hyperband search include $\{(500, 3), (700, 3), (500, 2), (900, 2), (1000, 2), (1050, 2)\}$. As shown in the figure, our approach outperforms Hyperband search in terms

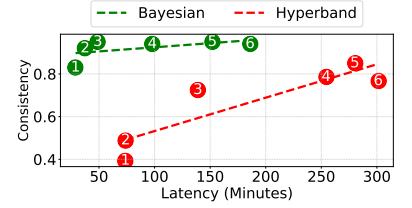


Fig. 6: Bayesian search vs. Hyperband search

of both tuning latency and consistency score. We observe that our approach achieves the peak consistency score of 0.95 for b_3 , which took 48.9 minutes, while the Hyperband search reached its maximum consistency score of 0.85 for h_5 , taking 280.15 minutes. Therefore, our approach outperforms Hyperband by $5.7\times$ in terms of tuning latency, while the peak consistency score achieved by our approach is 11.76% better than the best consistency score of Hyperband search.

VIII. CONCLUSIONS

The work is motivated by real-world requirements for a flexible and parameterized synthetic identity document generation framework from US General Services Administration and US Department of Homeland Security, where privacy regulations cause a lack of data for evaluating vendors' software. We proposed IDSPACE, a novel model-guided synthetic data generation approach, to address the shortage of accessible real data for trustworthy, reliable, comprehensive, and flexible evaluation of existing fraud detection models. Furthermore, our model-guided optimization-based strategy ensures that synthetic data can be aligned with the target domain using only a small set of real samples, significantly reducing the costs and dependency on sensitive real documents for both evaluation and training. Empirical evidence shows that IDSPACE improves evaluation consistency by 15 – 45% over baselines including CycleGAN, diffusion inpainting, and non-guided optimization, such as IDNet, using only 2 to 40 real samples, while improving training accuracy by up to 9% and SSIM similarity with the target domain by 10%, compared to IDNet.

IX. ACKNOWLEDGMENTS

This work was supported by the National Science Foundation under CAREER Award No. 2144923 and the U.S. Department of Homeland Security under Grant Award Number 17STQAC00001-07-00.

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